Precomposition with an essentially surjective functor is faithful

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'ct:esosurj-postcomp-faithful' (source lines 1376-1378, Lemma 9.9.1); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The original number of the counted result is not present in the label-number dump, so it was determined by counting the theorem-class environments of section 9 of the pinned source: the counted lemma is the first of that section, and the immediately following labelled result of the same section (ct:esofull-precomp-ff) is dump-confirmed as 9.9.2, which agrees with the count.

Editorial note. The retained counter values (chapter 9, section 9, theorem offset 0) make the counted lemma print as Lemma 9.9.1.

Editorial note. No context span is retained: the statement and proof use no label from outside the retained range.

Editorial scope. One original result of the HoTT Book, counted once: precomposition with an essentially surjective functor is faithful (Lemma 9.9.1).

Allowed standard prerequisites (editorial).

- Faithful functor. A functor is faithful when its action on each hom-type is injective.. It is the conclusion: precomposition with H must be injective on hom-types of functor precategories.
- Essentially surjective functor. H is essentially surjective when for every b there merely exists an a with H a = b.. It lets one assume a preimage and an isomorphism at source line 1382.
- Naturality of a natural transformation. A natural transformation commutes with the action of functors on morphisms.. It is what turns the middle equality of the displayed chain into the conjugation formula.

Lemma 9.9.1. *If A, B, C are precategories and $H : A \to B$ is an essentially surjective functor, then $(- \circ H) : C^B \to C^A$ is faithful.*

Proof. Let $F, G : B \to C$, and $\gamma, \delta : F \to G$ be such that $\gamma H = \delta H$; we must show $\gamma = \delta$. Thus let $b : B$; we want to show $\gamma_b = \delta_b$. This is a mere proposition, so since H is essentially surjective, we may assume given an $a : A$ and an isomorphism $f : Ha \cong b$. But now we have

$$\gamma_b = G(f) \circ \gamma_{Ha} \circ F(f^{-1}) = G(f) \circ \delta_{Ha} \circ F(f^{-1}) = \delta_b. \qquad \square$$